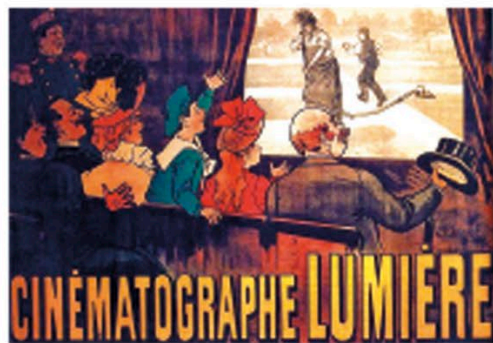


1895 ▶ 1900

1895

Cinema arrives

In Paris, France, 33 people formed the first paying cinema audience as they watched 10 short movies projected onto a screen by a cinématographe. This device, invented by French brothers Auguste and Louis Lumière, projected 16 photographs per second, giving the illusion of movement. The photos were taken on a 1.377-in- (35-mm-) wide film.



Poster advertising the first cinema show that people paid to see



Thomson using his cathode ray tube

1897

Discovery of electrons

While studying cathode rays (electric light tubes, see p.148), English scientist J. J. Thomson discovered electrons. These tiny particles orbit the center of atoms and carry a negative electrical charge. This was the first step toward understanding the structure of atoms. Thomson won the 1906 Nobel Prize in Physics for this discovery.

1897

World's largest telescope

Astronomers began using the world's largest refracting telescope (one that uses a lens rather than a mirror) at the Yerkes Observatory in Wisconsin. It has a 40-in- (102-cm-) diameter glass lens to gather in light from distant stars, planets, and galaxies. The telescope was used to discover the spiral shape of the Milky Way in 1951.



1895

1895 DISCOVERY OF X-RAYS



Röntgen's first X-ray image shows the bones and wedding ring on the hand of his wife, Anna.

While experimenting with electric light tubes, German physicist Wilhelm Conrad Röntgen discovered mysterious waves of energy that passed through flesh and paper, but not other materials such as bone and metals. He named them "X-rays" because "X" is used in mathematics to describe an unknown. Röntgen took the first X-ray images.

X-ray radiation

Early X-ray machines were basic and patients had to stand still for a long time for images to be captured. With little understanding of the potential dangers, staff and patients were often harmed by repeated or prolonged overexposure to these rays. With controlled and screened doses of radiation, X-rays have now become vital to detect bone fractures, lung problems, and foreign objects inside the body.



Röntgen using an early X-ray machine on a boy